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# MTH007: Ch 1.1 System of Linear Equations

**Year: 2022/23 Name: _____ ID: _____**

1. Solve the following system by elementary row operations

$$\begin{aligned} x_1 - 3x_2 + 4x_3 &= -4 \\ 3x_1 - 7x_2 + 7x_3 &= -8 \\ -4x_1 + 6x_2 - x_3 &= 7 \end{aligned}$$

Solution:

The augmented matrix is $\begin{bmatrix} 1 & -3 & 4 & -4 \\ 3 & -7 & 7 & -8 \\ -4 & 6 & -1 & 7 \end{bmatrix}$.

Replace $R_2$ by $R_2 + (-3)R_1$ and replace $R_3$ by $R_3 + (4)R_1$ we have:

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 3 & -7 & 7 & -8 \\ -4 & 6 & -1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ 0 & -6 & 15 & -9 \end{bmatrix}.$$

Then replace $R_3$ by $R_3 + (3)R_2$, and we have

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

The above reduced form indicates that the original linear system is inconsistent, because the last row would require that $0 = 3$ if there were a solution. Therefore, the solution set is empty.

2. Find an equation involving $g$, $h$ and $k$ that makes this augmented matrix corresponding to a consistent system:

$$\begin{pmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & k \end{pmatrix}.$$

Solution:
Firstly, the augmented matrix can be reduced to

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & k \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & -3 & 5 & k+2g \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & 0 & 0 & k+2g+h \end{bmatrix}.$$

Let $b$ denote the number $k+2g+h$. Then the third equation represented by the augmented matrix above is $0 = b$. This equation is consistent if and only if $b$ is zero. So the original system has a solution if and only if $k+2g+h = 0$ .

3. Find the elementary row operation that transforms the first matrix into the second, and then find the reverse row operation that transforms the second matrix into the first.

$$(1).\ \begin{pmatrix} 1 & 3 & -4 \\ 0 & -2 & 6 \\ 0 & -5 & 9 \end{pmatrix} \rightleftharpoons \begin{pmatrix} 1 & 3 & -4 \\ 0 & 1 & -3 \\ 0 & -5 & 9 \end{pmatrix};\ (2).\ \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 4 & -1 & 3 & -6 \end{pmatrix} \rightleftharpoons \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 0 & 7 & -1 & -6 \end{pmatrix}$$

Solution:
(1). Multiply $\mathrm{R}_2$ by $-1/2$; multiply $\mathrm{R}_2$ by $-2$.
(2). Replace $\mathrm{R}_3$ by $\mathrm{R}_3 + (-4)\mathrm{R}_1$; replace $\mathrm{R}_3$ by $\mathrm{R}_3 + (4)\mathrm{R}_1$

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# MTH007: Ch 1.2 Row Reduction to Echelon Forms

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1. Find the general solutions of the system whose augmented matrices are given by

$$\begin{pmatrix} 1 & 2 & -5 & -6 & 0 & -5 \\ 0 & 1 & -6 & -3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Solution:
Firstly, we reduce the above augmented matrix into the reduced echelon form

$$\begin{pmatrix} 1 & 2 & -5 & -6 & 0 & -5 \\ 0 & 1 & -6 & -3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 7 & 0 & 0 & -9 \\ 0 & 1 & -6 & -3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The corresponding linear system is

$$\begin{cases} x_1 + 7x_3 = -9 \\ x_2 - 6x_3 - 3x_4 = 2 \\ x_5 = 0 \\ 0 = 0. \end{cases}$$

Choosing $x_1, x_2, x_5$ as basic variables and $x_3$, $x_4$ as free variables, we have the general solutions

$$\begin{cases} x_1 = -7x_3 - 9 \\ x_2 = 6x_3 + 3x_4 + 2 \\ x_5 = 0 \\ x_3 \text{ is } free \\ x_4 \text{ is } free \end{cases}$$

Let $x_3 = s$ and $x_4 = t$. It follows that

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -9 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -7 \\ 6 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 3 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \ s, t \in \mathbb{R}.$$

2. Choose $k$ and $h$ such that the system $\begin{array}{c} x_1 + hx_2 = 2 \\ 4x_1 + 8x_2 = k \end{array}$ has

   (a) no solution

   (b) a unique solution

   (c) many solutions

Solution:
From

$$\begin{bmatrix} 1 & h & 2 \\ 4 & 8 & k \end{bmatrix} \sim \begin{bmatrix} 1 & h & 2 \\ 0 & 8-4h & k-8 \end{bmatrix},$$

we know that:
When $h = 2$ and $k \neq 8$, the augmented column is a pivot column, and the system is inconsistent.
When $h \neq 2$, the system is consistent and has a unique solution. There are no free variables.
When $h = 2$ and $k = 8$, the system is consistent and has infinitely many solutions.

3. A system of linear equations with more equations than unknowns is sometimes called an *overdetermined system*. Can such a system be consistent ? Illustrate your answer with a specific system of three equations in two unknowns.

Solution:
Yes, a system of linear equations with more equations than unknowns can be consistent.

Example: $\begin{aligned} x_1 + x_2 &= 3 \\ x_1 - 2x_2 &= 0. \\ 2x_1 - x_2 &= 3 \end{aligned}$

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# MTH007: Ch 1.3 Vector Equations

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1. Let $A = \begin{pmatrix} 1 & 0 & -4 \\ 0 & 0 & -2 \\ -2 & 6 & 3 \end{pmatrix}$ and $b = \begin{pmatrix} 4 \\ 1 \\ -4 \end{pmatrix}$. Denote the columns of $A$ by $a_1$, $a_2$, $a_3$, and let $W = \text{span}\{a_1, a_2, a_3\}$.

   (a) Is $b$ in $\{a_1, a_2, a_3\}$ ? How many vectors are in $\{a_1, a_2, a_3\}$?

   (b) Is $b$ in $W$ ? How many vectors are in $W$.

   (c) Show that $a_1$ is in $W$.

Solution:
(a). There are only three vectors in the set $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ , and $\mathbf{b}$ is not one of them.
(b). There are infinitely many vectors in $W = \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ . To determine if $\mathbf{b}$ is in $W$, we solve

$$x_1 \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} + x_3 \begin{pmatrix} -4 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ -4 \end{pmatrix}$$

From

$$\begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 0 & -2 & 1 \\ -2 & 6 & 3 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 0 & -2 & 1 \\ 0 & 6 & -5 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 6 & -5 & 4 \\ 0 & 0 & -2 & 1 \end{bmatrix}$$

we know that the augmented matrix is consistent, so $b \in W$.
(c). From $\mathbf{a}_1 = 1\mathbf{a}_1 + 0\mathbf{a}_2 + 0\mathbf{a}_3$, we know that $\mathbf{a}_1$ is in the space spanned by $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$.

2. Construct a $3 \times 3$ matrix $A$, with nonzero entries and a vector $b \in \mathbb{R}^3$ such that $b$ is **not** in the set spanned by the columns of $A$.

Solution:
Example:

$$A = \begin{pmatrix} 1 & 3 & 4 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \; b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

3. Let $v_1, \cdots, v_k$ be points in $\mathbb{R}^3$ and suppose that for $j = 1, \cdots, k$, an object with mass $m_j$ is located at point $v_j$. Physicists call such objects point masses. The total mass of the system point masses is

$$m = m_1 + \cdots + m_k.$$

The center of gravity (or center of mass) of the system is

$$\bar{v} = \frac{1}{m}(m_1 v_1 + \cdots + m_k v_k).$$

Compute the center of the gravity of the system consisting of the following point masses. Determine if $\bar{v}$ is in $Span\{v_1, \cdots, v_k\}$ ?

| points | mass |
|---|---|
| $v_1 = (5, -4, 3)$ | 2g |
| $v_2 = (4, 3, -2)$ | 5g |
| $v_3 = (-4, -3, -1)$ | 2g |
| $v_4 = (-9, 8, 6)$ | 1g |

Solution:
The total mass is $2 + 5 + 2 + 1 = 10$. So

$$\mathbf{v} = (2\mathbf{v}_1 + 5\mathbf{v}_2 + 2\mathbf{v}_3 + \mathbf{v}_4)/10.$$

That is,

$$\mathbf{v} = \frac{1}{10}\left(2\begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix} + 5\begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix} + 2\begin{bmatrix} -4 \\ -3 \\ -1 \end{bmatrix} + \begin{bmatrix} -9 \\ 8 \\ 6 \end{bmatrix}\right) = \frac{1}{10}\begin{bmatrix} 10 + 20 - 8 - 9 \\ -8 + 15 - 6 + 8 \\ 6 - 10 - 2 + 6 \end{bmatrix} = \begin{bmatrix} 1.3 \\ 0.9 \\ 0 \end{bmatrix}$$

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# MTH007: Ch 1.4 The Matrix Equation $Ax = b$

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1. Let $A = \begin{pmatrix} 2 & -1 \\ -6 & 3 \end{pmatrix}$ and $b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$.

   - Show that the equation $Ax = b$ does not have a solution for all possible $b$,
   - Describe the set of all $b$ for which $Ax = b$ does have a solution.

Solution:
The augmented matrix for $Ax = b$ is

$$\begin{bmatrix} 2 & -1 & b_1 \\ -6 & 3 & b_2 \end{bmatrix},$$

which is row equivalent to

$$\begin{bmatrix} 2 & -1 & b_1 \\ 0 & 0 & b_2 + 3b_1 \end{bmatrix}.$$

From the above reduced form we can see that the linear system has no solution if $b_2 + 3b_1 \neq 0$. For example when $b = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$.
The set of $b$ for which the equation is consistent is a line through the origin: the set of all points $(b_1, b_2)$ satisfying $b_2 = -3b_1$.

2. Let $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$. Explain whether $\{v_1, v_2, v_3\}$ span $\mathbb{R}^4$ ?

Solution:

Row reduce the matrix $\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix}$ to determine whether it has a pivot in each row.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix $\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix}$ does not have a pivot in each row, so the columns of the matrix do not span $\mathbb{R}^4$ , by Theorem 4. That is, $\{v_1, v_2, v_3\}$ does not span $\mathbb{R}^4$.

Another approach:

We can construct a counter-example:

Take a vector $b = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$ in $\mathbb{R}^4$. We can show that the vector $b = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$ can not be written as a linear combination of $\{v_1, v_2, v_3\}$. (You need to verify that the relevant system had no solution.)

Another approach: since the matrix has three columns and each column can have at most one pivot, there are at most three pivots in the matrix, fewer than four, the number of rows. By Theorem 4d, they cannot span $\mathbb{R}^4$.

3. Let $A$ be a $3 \times 4$ matrix. Let $y_1$ and $y_2$ be vectors in $\mathbb{R}^3$ and $w = y_1 + y_2$. Suppose

$$y_1 = Ax_1, \quad y_2 = Ax_2$$

for some vectors $x_1$ and $x_2$ in $\mathbb{R}^4$. Show that $Ax = w$ is consistent.

Solution:
Given $A\mathbf{x}_1 = \mathbf{y}_1$ and $A\mathbf{x}_2 = \mathbf{y}_2$, you are asked to show that the equation $A\mathbf{x} = \mathbf{w}$ has a solution, where $\mathbf{w} = \mathbf{y}_1 + \mathbf{y}_2$.
Observe that $\mathbf{w} = A\mathbf{x}_1 + A\mathbf{x}_2$ and use Theorem 5 $(\mathbf{a})$ with $\mathbf{x}_1$ and $\mathbf{x}_2$ in place of $\mathbf{u}$ and $\mathbf{v}$ , respectively. That is, $\mathbf{w} = A\mathbf{x}_1 + A\mathbf{x}_2 = A(\mathbf{x}_1 + \mathbf{x}_2)$. So the vector $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$ is a solution of $\mathbf{w} = A\mathbf{x}$

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# MTH007: Ch 1.5 Solution Sets of Linear Systems

**Year: 2022/23 Name: _____ ID: _____**

1. Determine if the system has a nontrivial solution.

$$\begin{aligned} x_1 - 3x_2 + 7x_3 &= 0 \\ -2x_1 + x_2 - 4x_3 &= 0 \\ x_1 + 2x_2 + 9x_3 &= 0 \end{aligned}$$

Solution:
The augumented matrix

$$\begin{bmatrix} 1 & -3 & 7 & 0 \\ -2 & 1 & -4 & 0 \\ 1 & 2 & 9 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 7 & 0 \\ 0 & -5 & 10 & 0 \\ 0 & 5 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 7 & 0 \\ 0 & -5 & 10 & 0 \\ 0 & 0 & 12 & 0 \end{bmatrix}$$

The linear system corresponding to the above echelon form is

$$\begin{aligned} x_1 - 3x_2 + 7x_3 &= 0 \\ -5x_2 + 10x_3 &= 0 \\ 12x_3 &= 0 \end{aligned}$$

and we obtain $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \mathbf{0}$. Therefore the system has only the trivial solution.

2. Describe the solutions of the following system in parametric vector form:

$$\begin{aligned} x_1 + 3x_2 - 5x_3 &= 4 \\ x_1 + 4x_2 - 8x_3 &= 7 \\ -3x_1 - 7x_2 + 9x_3 &= -6 \end{aligned}$$

Solution:
The row reduction process of the augmented matrix for the system is as follows:

$$\begin{bmatrix} 1 & 3 & -5 & 4 \\ 1 & 4 & -8 & 7 \\ -3 & -7 & 9 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -5 & 4 \\ 0 & 1 & -3 & 3 \\ 0 & 2 & -6 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -5 & 4 \\ 0 & 1 & -3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4 & -5 \\ 0 & 1 & -3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The corresponding linear system is

$$\begin{cases} x_1 + 4x_3 = -5 \\ x_2 - 3x_3 = 0 \\ 0 = 0 \end{cases}$$

Thus $x_1 = -5 - 4x_3$, $x_2 = 3 + 3x_3$ and $x_3$ is free.

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -5 - 4x_3 \\ 3 + 3x_3 \\ x_3 \end{pmatrix} = \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -4 \\ 3 \\ 1 \end{pmatrix}, \ x_3 \in \mathbb{R}.$$

3. Given $A = \begin{pmatrix} -2 & -6 \\ 7 & 21 \\ -3 & -9 \end{pmatrix}$, find one nontrivial solution of $Ax = 0$ by inspection.

Solution:
Look at

$$x_1 \begin{bmatrix} -2 \\ 7 \\ -3 \end{bmatrix} + x_2 \begin{bmatrix} -6 \\ 21 \\ -9 \end{bmatrix}$$

and notice that the second column is 3 times the first.
So suitable values for $x_1$ and $x_2$ would be 3 and -1 respectively. (Another pair would be 6 and -2, etc). Thus $x = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ satisfies $A\mathbf{x} = 0$

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# MTH007: Ch 1.7 Linear Independence

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1. Find the value(s) of $h$ for which the vectors

$$v_1 = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \; v_2 = \begin{pmatrix} -5 \\ 7 \\ 8 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ h \end{pmatrix}$$

are linearly dependent.

Solution: By the definition of linear dependence, the linear system $c_1v_1 + c_2v_2 + c_3v_3 = 0$ for $c_i \in \mathbb{R}$, $i = 1, 2, 3$ must have nontrivial solutions. This is equivalent to that any row echelon form of the matrix $V = (v_1, v_2, v_3)$ has at least one row of all zeros. The row reduction process can be demonstrated as follows:

$$\begin{pmatrix} 1 & -5 & 1 \\ -1 & 7 & 1 \\ 3 & 8 & h \end{pmatrix}$$

(Row 2 + Row 1; Row 3 + (-3) Row 1 ) $\rightarrow$

$$\begin{pmatrix} 1 & -5 & 1 \\ 0 & 2 & 2 \\ 0 & 23 & -3+h \end{pmatrix}$$

(0.5 $\times$ Row 2) $\rightarrow$

$$\begin{pmatrix} 1 & -5 & 1 \\ 0 & 1 & 1 \\ 0 & 23 & -3+h \end{pmatrix}$$

(Row 3 + (-23) Row 2; Row 1 + (5) Row 2) $\rightarrow$

$$\begin{pmatrix} 1 & 0 & 6 \\ 0 & 1 & 1 \\ 0 & 0 & -26+h \end{pmatrix}$$

Therefore, $-26 + h$ must vanish, i.e. $h = 26$.

2. Let

$$v_1 = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} -3 \\ 9 \\ -6 \end{pmatrix}, v_3 = \begin{pmatrix} 5 \\ -7 \\ h \end{pmatrix}$$

- For what values of $h$ is $v_3$ in $Span\{v_1, v_2\}$;
- For what values of $h$ is $\{v_1, v_2, v_3\}$ linearly dependent.

Solution:

• The first question is equivalent to asking whether $c_1v_1 + c_2v_2 = v_3$ has a solution for $c_1, c_2 \in \mathbb{R}$. We do the row reduction of the augmented matrix $(v_1, v_2, v_3)$:

$$\begin{pmatrix} 1 & -3 & 5 \\ -3 & 9 & -7 \\ 2 & -6 & h \end{pmatrix}$$

(Row 2 + (3) Row 1; Row 3 + (-2) Row 1)$\rightarrow$

$$\begin{pmatrix} 1 & -3 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & -10+h \end{pmatrix}$$

Note that the second row indicates a contradiction since $0 \neq 8$. Hence, there is no solution for the linear algebraic system $c_1v_1 + c_2v_2 = v_3$. That is, for any value of $h$, we have that $v_3$ is not in $Span\{v_1, v_2\}$.

• The second question is equivalent to asking whether $c_1v_1 + c_2v_2 + c_3v_3 = 0$ has nontrivial solutions. Continue the row reduction process for the matrix

$$\begin{pmatrix} 1 & -3 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & -10+h \end{pmatrix}$$

($\frac{1}{8}\times$ Row 2)$\rightarrow$

$$\begin{pmatrix} 1 & -3 & 5 \\ 0 & 0 & 1 \\ 0 & 0 & -10+h \end{pmatrix}$$

(Row 3 + (10-h) Row 2)$\rightarrow$

$$\begin{pmatrix} 1 & -3 & 5 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Therefore, for any value of $h$, $\{v_1, v_2, v_3\}$ are linearly dependent.

Remark: Observe that $v_2 = -3v_1$. It immediately follows that $\{v_1, v_2\}$ are linearly dependent. So $Span\{v_1, v_2\} = Span\{v_1\}$, and $\{v_1, v_2, v_3\}$ are linearly dependent. Note that $\{v_1, v_3\}$ are linearly independent. Hence, $v_3$ can never be in $Span\{v_1\}$.

3. Given

$$A = \begin{pmatrix} 2 & 3 & 5 \\ -5 & 1 & -4 \\ -3 & -1 & -4 \\ 1 & 0 & 1 \end{pmatrix},$$

observe that the third column is the sum of the first two columns. Find a nontrivial solution of $Ax = 0$.

Solution: Let $A$ be partitioned as columns $(a_1, a_2, a_3)$. Based on the observation, we have $a_1 + a_2 = a_3$. In other words, $a_1 + a_2 - a_3 = 0$. Note that $Ax = 0$ can be written as $x_1a_1 + x_2a_2 + x_3a_3 = 0$. Hence, $Ax = 0$ has a nontrivial solution $x = (1, 1, -1)^T$.

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# MTH007: Ch 2.1 Matrix Operations

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1. If $A = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$ and $AB = \begin{pmatrix} -1 & 2 & -1 \\ 6 & -9 & 3 \end{pmatrix}$, determine the first and the second columns of $B$.

Solution:

Let $B$ be partitioned into its columns $B = (b_1, b_2, b_3)$, $b_i \in \mathbb{R}^2$. We have $AB = A(b_1, b_2, b_3) = (Ab_1, Ab_2, Ab_3)$. We know that

$$Ab_1 = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix} b_1 = \begin{pmatrix} -1 \\ 6 \end{pmatrix}.$$

Similarly, we need to solve also $Ab_2 = (2, -9)^T$. The row reduction can be done together:

$$\left(\begin{array}{cc|c|c} 1 & -2 & -1 & 2 \\ -2 & 5 & 6 & -9 \end{array}\right)$$

(Row 2 + (2) Row 1) $\rightarrow$

$$\left(\begin{array}{cc|c|c} 1 & -2 & -1 & 2 \\ 0 & 1 & 4 & -5 \end{array}\right)$$

(Row 1 + (2) Row 1) $\rightarrow$

$$\left(\begin{array}{cc|c|c} 1 & 0 & 7 & -8 \\ 0 & 1 & 4 & -5 \end{array}\right)$$

Hence, $b_1 = (7, 4)^T$ and $b_2 = (-8, -5)^T$.

2. Let $u = \begin{pmatrix} -2 \\ 3 \\ -4 \end{pmatrix}$ and $v = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. Compute $u^T v$, $v^T u$, $uv^T$ and $vu^T$.

Solution:

$$u^T v = (-2, 3, -4) \begin{pmatrix} a \\ b \\ c \end{pmatrix} = -2a + 3b - 4c.$$

$$v^T u = (u^T v)^T = -2a + 3b - 4c.$$

$$uv^T = \begin{pmatrix} -2 \\ 3 \\ -4 \end{pmatrix} (a, b, c) = \begin{pmatrix} -2a & -2b & -2c \\ 3a & 3b & 3c \\ -4a & -4b & -4c \end{pmatrix}.$$

$$vu^T = (uv^T)^T = \begin{pmatrix} -2a & 3a & -4a \\ -2b & 3b & -4b \\ -2c & 3c & -4c \end{pmatrix}.$$

3. Suppose $AD = I_m$ (the $m \times m$ identity matrix). Show that for any vector $b \in \mathbb{R}^m$, the equation $Ax = b$ is consistent.

Solution: From the given conditions, we have for any $b \in \mathbb{R}^m$ that

$$b = I_m b = ADb = A(Db).$$

That is, $Ax = b$ has at least one solution $x = Db$. Hence, $Ax = b$ is consistent.

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# MTH007: Ch 2.2 The inverse of a Matrix

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1. Let $A = \begin{pmatrix} 1 & 2 \\ 5 & 12 \end{pmatrix}$, $b_1 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$, $b_2 = \begin{pmatrix} 1 \\ -5 \end{pmatrix}$, $b_3 = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$ and $b_4 = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$.

   - Find $A^{-1}$, and use it to solve the four equations $Ax = b_i$, $i = 1, 2, 3, 4$.
   - The four equations in the last question can be solved by the same set of row operations, since the coefficient matrix is the same in each case. Solve the four equations in the last question by row reducing the augmented matrix $[A\ b_1\ b_2\ b_3]$.

Solution:

• Note that $AA^{-1} = I$. Denote the columns of $A^{-1}$ by $A^{-1} = (u_1, u_2)$, $u_1, u_2 \in \mathbb{R}^2$. We see that $Au_i = e_i$ for $i = 1, 2$. So finding $A^{-1}$ is equivalent to solving $Au_i = e_i$, $i = 1, 2$. We do the row reduction for the augmented matrix $(A, I)$ as follows.

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 5 & 12 & 0 & 1 \end{array}\right)$$

(Row 2 + (-5) Row 1) $\rightarrow$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 2 & -5 & 1 \end{array}\right)$$

(0.5$\times$ Row 2) $\rightarrow$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -2.5 & 0.5 \end{array}\right)$$

(Row 1 + (-2) Row 2) $\rightarrow$

$$\left(\begin{array}{cc|cc} 1 & 0 & 6 & -1 \\ 0 & 1 & -2.5 & 0.5 \end{array}\right)$$

Therefore,

$$A^{-1} = \begin{pmatrix} 6 & -1 \\ -2.5 & 0.5 \end{pmatrix}$$

The solution of $Ax = b_1$ is thus given by

$$x = A^{-1}b_1 = \begin{pmatrix} 6 & -1 \\ -2.5 & 0.5 \end{pmatrix} \begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -9 \\ 4 \end{pmatrix}.$$

Similarly, for $Ax = b_2$ we have

$$x = \begin{pmatrix} 6 & -1 \\ -2.5 & 0.5 \end{pmatrix} \begin{pmatrix} 1 \\ -5 \end{pmatrix} = \begin{pmatrix} 11 \\ -5 \end{pmatrix}.$$

For $Ax = b_3$ we get

$$x = \begin{pmatrix} 6 & -1 \\ -2.5 & 0.5 \end{pmatrix} \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 6 \\ -2 \end{pmatrix}.$$

For $Ax = b_4$ we obtain

$$x = \begin{pmatrix} 6 & -1 \\ -2.5 & 0.5 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 13 \\ -5 \end{pmatrix}.$$

• The row reduction for $(A, b_1, b_2, b_3)$ should give the same solutions.

$$\left(\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 5 & 12 & 3 & -5 & 6 & 5 \end{array}\right)$$

(Row 2 + (-5) Row 1) $\rightarrow$

$$\left(\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 2 & 8 & -10 & -4 & -10 \end{array}\right)$$

(0.5$\times$ Row 2) $\rightarrow$

$$\left(\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{array}\right)$$

(Row 1 + (-2) Row 2) $\rightarrow$

$$\left(\begin{array}{cc|cccc} 1 & 0 & -9 & 11 & 6 & 13 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{array}\right)$$

The last four columns are the solutions for $Ax = b_i$, $i = 1, 2, 3, 4$.

2. Suppose $AB = AC$, where $B$ and $C$ are $n \times p$ matrices and $A$ is invertible. Show that $B = C$. Is this true when $A$ is not invertible ?

Solution: • Since $A$ is invertible, we hit the equation $AB = AC$ on the left with $A^{-1}$ to find $A^{-1}AB = A^{-1}AC$. Hence, $(A^{-1}A)B = (A^{-1}A)C$, i.e. $B = C$. (Remark: We have used the association law for the matrix multiplication. But the commutation law is in general not true for the matrix multiplication, i.e. $AB$ is not always equal to $BA$ except when $A$ and $B$ have special relations.)

• If $A \in \mathbb{R}^{n\times n}$ is not invertible, then $Ax = 0$ has nontrivial solutions. Therefore, $AB = AC$ or $A(B - C) = 0$ does not imply $B - C = 0$.

Further remark when $A$ is not a square matrix. If $A \in \mathbb{R}^{m\times n}$ with $m < n$, then $Ax = 0$ must have nontrivial solutions because the number of pivots in a row echlon form is less than $n$.

If $A \in \mathbb{R}^{m\times n}$ with $m > n$, we can calculate the echelon form of $A$. If the echelon form of $A$ has exactly $n$ nonzero rows (or, equivalently, $n$ pivots), then $Ax = 0$ has only the trivial solution and henceforce $B = C$. (Remark: In this case, $A$ has the left inverse $(A^TA)^{-1}A^T$ such that $\left((A^TA)^{-1}A^T\right)A = I_n$.) If the echelon form of $A$ has less than $n$ nonzero rows, then $Ax = 0$ has nontrivial solutions.

3. Find the inverse of the matrix

$$A = \begin{pmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{pmatrix}.$$

Solution:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right)$$

(Row 2 + (3) Row 1; Row 3 + (-2) Row 1) $\rightarrow$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{array}\right)$$

(Row 3 + (3) Row 2) $\rightarrow$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right)$$

((1/2) Row 3) $\rightarrow$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 7/2 & 3/2 & 1/2 \end{array}\right)$$

(Row 2 + (2) Row 3; Row 1 + (2) Row 3) $\rightarrow$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & 7/2 & 3/2 & 1/2 \end{array}\right)$$

The right part is $A^{-1}$.

Remark: In all the solutions, we need the row reduction operations. We have used the scaling and adding operations. It is just a coincidence that we did not use the operation of interchanging two rows. Nevertheless, interchanging two rows can be very useful or even necessary. For example, consider the folowing matrix (modifed from Problem 3)

$$\begin{pmatrix} 0 & 1 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{pmatrix}.$$

The (1,1)-entry cannot be used as the pivot for the first column because it is zero!

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# MTH007: Ch 2.3 Characterizations of Invertible Matrices

**Year: 2022/23 Name: _____ ID: _____**

1. If $A$ is invertible, show that the columns of $A^{-1}$ are linearly independent.

Solution:
We prove it by finding a contradiction.
Suppose the columns of $A^{-1}$ are linearly dependent. Then there exists a nonzero vector $e$ such that

$$A^{-1}e = 0,$$

from which we have

$$A(A^{-1}e) = e = 0.$$

This is contradicted with the assumption that $e$ is a nonzero vector. Thus the columns of $A^{-1}$ are linearly independent.

2. Determine whether the matrix $A = \begin{pmatrix} -1 & -3 & 0 & 1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{pmatrix}$ is invertible.

Solution:

$$\begin{pmatrix} -1 & -3 & 0 & 1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{pmatrix}$$

((-1) Row 1) $\rightarrow$

$$\begin{pmatrix} 1 & 3 & 0 & -1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{pmatrix}$$

(Row 2 + (-3) Row 1; Row 3 + (2) Row 1) $\rightarrow$

$$\begin{pmatrix} 1 & 3 & 0 & -1 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -1 & 2 & 1 \end{pmatrix}$$

(Interchange Row 3 and Row 4) $\rightarrow$

$$\begin{pmatrix} 1 & 3 & 0 & -1 \\ 0 & -4 & 0 & 0 \\ 0 & -1 & 2 & 1 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$

((-1/4 Row 2)) $\rightarrow$

$$\begin{pmatrix} 1 & 3 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 2 & 1 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$

(Row 3 + Row 2)) $\rightarrow$

$$\begin{pmatrix} 1 & 3 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$

(Row 4 + (-3/2) Row 3)) $\rightarrow$

$$\begin{pmatrix} 1 & 3 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -3/2 \end{pmatrix}$$

In the echelon form, every column has a pivot. Moreover, $A$ is a square matrix. Hence, $A$ is invertible.

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# MTH007: Ch 2.4 Partitioned Matrices*

**Year: 2022/23 Name: _____ ID: _____**

1. The inverse of $\begin{pmatrix} I & 0 & 0 \\ C & I & 0 \\ A & B & I \end{pmatrix}$ is $\begin{pmatrix} I & 0 & 0 \\ Z & I & 0 \\ X & Y & I \end{pmatrix}$. Find $X$, $Y$ and $Z$.

Solution:

$$\begin{bmatrix} I & 0 & 0 \\ C & I & 0 \\ A & B & I \end{bmatrix} \begin{bmatrix} I & 0 & 0 \\ Z & I & 0 \\ X & Y & I \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

By block matrix multiplication we have

$$\begin{bmatrix} I & 0 & 0 \\ C & I & 0 \\ A & B & I \end{bmatrix} \begin{bmatrix} I & 0 & 0 \\ Z & I & 0 \\ X & Y & I \end{bmatrix}$$

$$= \begin{bmatrix} II + 0Z + 0X & I0 + 0I + 0Y & I0 + 00 + 0I \\ CI + IZ + 0X & C0 + II + 0Y & C0 + I0 + 0I \\ AI + BZ + IX & A0 + BI + IY & A0 + B0 + II \end{bmatrix}$$

$$= \begin{bmatrix} I & 0 & 0 \\ C + Z & I & 0 \\ A + BZ + X & B + Y & I \end{bmatrix}$$

$$= \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}.$$

Form $C + Z = 0$ we have $Z = -C$; from $B + Y = 0$, we have $Y = -B$; from $A + BZ + X = 0$, we have $X = -A - BZ = BC - A$.

2. Verify that $A^2 = I$ when $A = \begin{pmatrix} 1 & 0 \\ 3 & -1 \end{pmatrix}$. Then Partition of matrices to show that $M^2 = I$ when

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -3 & 1 \end{pmatrix}.$$

Solution: First we verify that $A^2 = I$ by direct matrix multiplication: $A^2 = \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 3-3 & 0+(-1)^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. For the second part, we take the advantage of block matrix multiplication

$$M^2 = \begin{bmatrix} A & 0 \\ I & -A \end{bmatrix} \begin{bmatrix} A & 0 \\ I & -A \end{bmatrix} = \begin{bmatrix} A^2 + 0 & 0 + 0 \\ A - A & 0 + (-A)^2 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

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# MTH007: Ch 2.8 Subspaces of $\mathbb{R}^n$

**Year: 2022/23 Name: _____ ID: _____**

1. Let $v_1 = \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix}$, $v_2 = \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}$, $v_3 = \begin{pmatrix} 0 \\ -6 \\ 3 \end{pmatrix}$ and $p = \begin{pmatrix} 1 \\ 14 \\ -9 \end{pmatrix}$. Determine if $p$ is in Col(A).

Solution:

$$\begin{bmatrix} -3 & -2 & 0 & 1 \\ 0 & 2 & -6 & 14 \\ 6 & 3 & 3 & -9 \end{bmatrix} - \begin{bmatrix} -3 & -2 & 0 & 1 \\ 0 & 2 & -6 & 14 \\ 0 & -1 & 3 & -7 \end{bmatrix} - \begin{bmatrix} -3 & -2 & 0 & 1 \\ 0 & 2 & -6 & 14 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Yes the augmented matrix $[A, p]$ corresponds to a consistent system, so $p$ is in $Col(A)$.

2. Let

$$A = \begin{pmatrix} 1 & 4 & 8 & 3 & -7 \\ -1 & 2 & 7 & 3 & 4 \\ -2 & 2 & 9 & 5 & 5 \\ 3 & 6 & 9 & -5 & -2 \end{pmatrix}.$$

Find a basis for $Col(A)$ and $Nul(A)$.

Solution: We first find the echelon form of $A$

$$A = \begin{bmatrix} 1 & 4 & 8 & -3 & -7 \\ -1 & 2 & 7 & 3 & 4 \\ -2 & 2 & 9 & 5 & 5 \\ 3 & 6 & 9 & -5 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 8 & 3 & -7 \\ 0 & 6 & 15 & 6 & -3 \\ 0 & 10 & 25 & 11 & -9 \\ 0 & -6 & -15 & -14 & 19 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 8 & 3 & -7 \\ 0 & 2 & 5 & 2 & -1 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -16 \end{bmatrix}.$$

So the basis for $Col(A)$ is

$$\begin{bmatrix} 1 \\ -1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \\ 5 \\ -5 \end{bmatrix}, \begin{bmatrix} -7 \\ 4 \\ 5 \\ -2 \end{bmatrix}.$$

To find the $Nul(A)$, we find the reduced (and augmented) echelon form of $Ax = 0$

$$\begin{bmatrix} A & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 0 & 0 \\ 0 & 1 & 2.5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{aligned} x_1 - 2x_3 &= 0 \\ x_2 + 2.5x_3 &= 0 \\ x_4 &= 0 \\ x_5 &= 0 \end{aligned}$$

The solution of $Ax = 0$ in parametric vector form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_3 \\ -2.5x_3 \\ x_3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -2.5 \\ 1 \\ 0 \\ 0 \end{bmatrix} k, \quad k \in \mathbb{R}.$$

The basis for $Nul(A)$ is $(2, -2.5, 1, 0, 0)^T$.

3. Construct a nonzero $3 \times 3$ matrix $A$ and a vector $b$ such that $b$ is not in $Col(A)$.

Solution: The easiest construction is to write a $3 \times 3$ matrix in echelon form that has only 2 pivots, and let b be any vector in $\mathbb{R}^3$ whose third entry is nonzero.

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# MTH007: Ch 2.9 Dimension and Rank

**Year: 2022/23 Name: _____ ID: _____**

1. Find a basis for the subspace spanned by

$$v_1 = \begin{pmatrix} 1 \\ -3 \\ 2 \\ -4 \end{pmatrix}, v_2 = \begin{pmatrix} -3 \\ 9 \\ -6 \\ 12 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ -1 \\ 4 \\ 2 \end{pmatrix}, v_4 = \begin{pmatrix} -4 \\ 5 \\ -3 \\ 7 \end{pmatrix}.$$

Then determine the dimension of the subspace.

Solution: Apply the row reductions to the matrix consisting of the four vectors as columns to the echelon from:

$$\begin{bmatrix} 1 & -3 & 2 & -4 \\ -3 & 9 & -1 & 5 \\ 2 & -6 & 4 & -3 \\ -4 & 12 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 0 & 5 & -7 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 10 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 0 & 5 & -7 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Columns 1,3 and 4 of the original matrix form a basis for $H$, so $\dim H = 3$

2. Suppose vectors $b_1, \cdots, b_p$ span a subspace $W$ and let $\{a_1, \ldots, a_q\}$ be any set in $W$ containing more than $p$ vectors. Let $B = [b_1, \cdots, b_p]$ and $A = [a_1, \cdots, a_q]$. Fill in the details of the following argument to show that $\{a_1, \ldots, a_q\}$ must be linearly dependent.

- Explain why for each vector $a_j$, there exists a vector $c_j \in \mathbb{R}^p$, such that $a_j = Bc_j$.
- Let $C = [c_1, \cdots, c_q]$. Explain why there exists a nonzero vector $u$ such that $Cu = 0$.
- Use $B$ and $C$ to show that $Au = 0$.

Solution:

- Start with $B = \begin{bmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_p \end{bmatrix}$ and$A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_q \end{bmatrix}$, where $q > p$. For $j = 1, ..., q$, the vector $a_j$ is in $W$. Since the columns of $B$ span $W$, the vector $a_j$ is in the column space of $B$. That is, $a_j = Bc_j$ for some vector $c_j$ of weights. Note that $c_j$ is in $\mathbb{R}^p$ because $B$ has $p$ columns.
- Let $C = [c_1, \cdots, c_q]$. Then $C$ is a $p \times q$ matrix with $p < q$, so $C$ has more columns than rows. By the rank theorem, $NulC$ cannot be $\{0\}$.
- From part (a) and the definition of matrix multiplication $A = [a_1, \cdots, a_q] = [Bc_1, \cdots, Bc_q] = BC$. From part (b) $Au = (BC)u = B(Cu) = B0 = 0$. Since $u$ is nonzero, the columns of $A$ are linearly dependent.

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# MTH007: Ch 3.1 Introduction to determinant

**Year: 2022/23 Name: _____ ID: _____**

1. Compute the determinant by cofactor expansions. At each step, choose a row or column that involves the least amount of computation.

$$\begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix}$$

Solution: First expand along either the second row or the second column. Using the second row,

$$\begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix} = (-1)^{2+3} \cdot 2 \begin{vmatrix} 4 & 0 & 3 & -5 \\ 7 & 3 & 4 & -8 \\ 5 & 0 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{vmatrix}$$

Now expand along the second column to find:

$$(-1)^{2+3} \cdot 2 \begin{vmatrix} 4 & 0 & 3 & -5 \\ 7 & 3 & 4 & -8 \\ 5 & 0 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{vmatrix} = -2 \begin{vmatrix} 4 & 3 & -5 \\ 5 & 2 & -3 \\ 0 & -1 & 2 \end{vmatrix}.$$

Now expand along either the first column or third row. The first column is used below.

$$-2(-1)^{2+2} \cdot 3 \begin{vmatrix} 4 & 3 & -5 \\ 5 & 2 & -3 \\ 0 & -1 & 2 \end{vmatrix} = -6 \left( (-1)^{1+1} \cdot 4 \begin{vmatrix} 2 & -3 \\ -1 & 2 \end{vmatrix} + (-1)^{2+1} \cdot 5 \begin{vmatrix} 3 & -5 \\ -1 & 2 \end{vmatrix} \right)$$

$$= (-6)(4(1) - 5(1)) = 6$$

2. Let $u = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

   - Compute the area of the parallelogram determined by $u$, $v$, $u + v$ and $\mathbf{0}$;
   - Compute the determinant of $[u, v]$.
   - Replace the first entry of $v$ by an arbitrary number $x$, and repeat the problem.
   - Draw a picture and explain your findings.

Solution:

- The area is 6, since the base of the parallelogram has length 3 and the height of the parallelogram is 2.
- The determinant is also $6$.
- By the same reasoning the area of the parallelogram determined by $u$, $v$ and $u + v$ and $0$ is also 6.
- The determinant of the matrix whose columns are those vectors which define the sides of the parallelogram adjacent to 0 is equal to the area of the parallelogram. All parallelograms with the same base and equal heights have the same area.

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# MTH007: Ch 3.2 Properties of determinant

**Year: 2022/23 Name: _____ ID: _____**

1. Find the determinant by row reduction to echelon form.

$$\begin{vmatrix} 1 & 3 & -1 & 0 & 2 \\ 0 & 2 & -4 & -2 & -6 \\ -2 & -6 & 2 & 3 & 10 \\ 1 & 5 & -6 & 2 & -3 \\ 0 & 2 & -4 & 5 & 9 \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 1 & 3 & -1 & 0 & 2 \\ 0 & 2 & -4 & -2 & -6 \\ -2 & -6 & 2 & 3 & 10 \\ 1 & 5 & -6 & 2 & -3 \\ 0 & 2 & -4 & 5 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -1 & 0 & 2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & 0 & 3 & 14 \\ 0 & 2 & -5 & 2 & -5 \\ 0 & 2 & -4 & 5 & 9 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 3 & -1 & 0 & 2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & 0 & 3 & 14 \\ 0 & 0 & -1 & 4 & 1 \\ 0 & 0 & 0 & 7 & 15 \end{vmatrix} = - \begin{vmatrix} 1 & 3 & -1 & 0 & 2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & -1 & 4 & 1 \\ 0 & 0 & 0 & 3 & 14 \\ 0 & 0 & 0 & 7 & 15 \end{vmatrix} = - \begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & -1 & 4 & 5 \\ 0 & 0 & 0 & 3 & 14 \\ 0 & 0 & 0 & 0 & -\frac{53}{3} \end{vmatrix}$$

$$= (-3)(-2)\frac{-53}{3} = -106$$

2. Suppose

$$\begin{vmatrix} a & b & c \\ d & e & f \\ h & i & j \end{vmatrix} = 7.$$

Find the value of the determinant

$$\begin{vmatrix} a & b & c \\ 2d+a & 2e+b & 2f+c \\ h & i & j \end{vmatrix}$$

by using the properties of determinant.

Solution:

$$\begin{vmatrix} a & b & c \\ 2d+a & 2e+b & 2f+c \\ h & i & j \end{vmatrix} = \begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ h & i & j \end{vmatrix} = 2\begin{vmatrix} a & b & c \\ d & e & f \\ h & i & j \end{vmatrix} = 2(7) = 14$$

3. Using the definition of determinant to prove that

$$det(A) = det(B) + det(C),$$

where $A = \begin{pmatrix} a_{11} & a_{12} & u_1 + v_1 \\ a_{21} & a_{22} & u_2 + v_2 \\ a_{31} & a_{32} & u_3 + v_3 \end{pmatrix}$, $B = \begin{pmatrix} a_{11} & a_{12} & u_1 \\ a_{21} & a_{22} & u_2 \\ a_{31} & a_{32} & u_3 \end{pmatrix}$ and $C = \begin{pmatrix} a_{11} & a_{12} & v_1 \\ a_{21} & a_{22} & v_2 \\ a_{31} & a_{32} & v_3 \end{pmatrix}$.

Solution: Compute det A by using a cofactor expansion down the third column:

$$\begin{aligned} \det A &= (u_1 + v_1) \det A_{13} - (u_2 + v_2) \det A_{23} + (u_3 + v_3) \det A_{33} \\ &= u_1 \det A_{13} - u_2 \det A_{23} + u_3 \det A_{33} + v_1 \det A_{13} - v_2 \det A_{23} + v_3 \det A_{33} \\ &= \det B + \det C \end{aligned}$$

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# MTH007: Ch 3.3 Cramer's rule

**Year: 2022/23 Name: _____ ID: _____**

1. Using Cramer's rule to compute the solutions of the following system:

$$\begin{aligned} x_1 + 3x_2 + x_3 &= 4 \\ -x_1 \quad\quad + 2x_3 &= 2 \\ 3x_1 + x_2 \quad\quad &= 2 \end{aligned}$$

Solution: The system is equivalent to $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} 1 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}$.

We compute

$$A_1(\mathbf{b}) = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix}, A_2(\mathbf{b}) = \begin{bmatrix} 1 & 4 & 1 \\ -1 & 2 & 2 \\ 3 & 2 & 0 \end{bmatrix}, A_3(\mathbf{b}) = \begin{bmatrix} 1 & 3 & 4 \\ -1 & 0 & 2 \\ 3 & 1 & 2 \end{bmatrix},$$

$$\det A = 15, \det A_1(\mathbf{b}) = 6, \det A_2(\mathbf{b}) = 12, \det A_3(\mathbf{b}) = 18,$$

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{6}{15} = \frac{2}{5}, x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = \frac{12}{15} = \frac{4}{5}, x_3 = \frac{\det A_3(\mathbf{b})}{\det A} = \frac{18}{15} = \frac{6}{5}.$$

2. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at $(1, 3, 0)$, $(-2, 0, 2)$ and $(-1, 3, -1)$.

Solution: The parallelepiped is determined by the columns of $A = \begin{bmatrix} 1 & -2 & -1 \\ 3 & 0 & 3 \\ 0 & 2 & -1 \end{bmatrix}$, so the volume of the parallelepiped is $|\det A| = |-18| = 18$.

3. Let $\mathbb{S}$ be the parallelogram determined by the vectors $b_1 = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $b_2 = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$. Let $A = \begin{pmatrix} 6 & -3 \\ -3 & 2 \end{pmatrix}$. Compute the area of the image of $\mathbb{S}$ under the mapping $x \mapsto Ax$.

Solution: since the parallelogram $S$ is determined by the columns of $\begin{bmatrix} -2 & -2 \\ 3 & 5 \end{bmatrix}$, the area of $S$ is $\left|\det \begin{bmatrix} -2 & -2 \\ 3 & 5 \end{bmatrix}\right| = |-4| = 4$. The matrix $A$ has $\det A = \begin{vmatrix} 6 & -3 \\ -3 & 2 \end{vmatrix} = 3$. By Theorem 10, the area of $T(S)$ is $|\det A| \{$ area of $S\} = 3 \cdot 4 = 12$. Alternatively, one may compute the vectors that determine the image, namely, the columns of $A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} -2 & -2 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} -21 & -27 \\ 12 & 16 \end{bmatrix}$. The determinant of this matrix is $-12$, so the area of the image is $12$.

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# MTH007: Ch 4.1 Vector space and subspace

**Year: 2022/23 Name: _____ ID: _____**

1. Let $F$ be a fixed $3 \times 2$ matrix, and let $H$ be the set of all matrices $A$ in $M_{2\times 4}$ with the property that $FA = 0$. Determine if H is a subspace of $M_{2\times 4}$.

Solution: The set $H$ is a subspace of $M_{2\times 4}$. The $2 \times 4$ zero matrix 0 is in $H$ because $F0 = 0$. If $A$ and $B$ are matrices in $H$, then $F(A + B) = FA + FB = 0 + 0 = 0$, so $A + B$ is in $H$. If $A$ is in $H$ and $c$ is a scalar, then $F(cA) = c(FA) = c0 = 0$, so $cA$ is in $H$.

2. Let $H$ and $K$ be subspaces of vector space $V$. The intersection of $H$ and $K$, written as $H \cap K$, is the set of $v$ in $V$ that belong to both $H$ and $K$. Show that $H \cap K$ is a subspace of $V$. Give an example in $R^2$ to show that the union of two subspaces is not, in general, a subspace.

Solution: Both $H$ and $K$ contain the zero vector of $V$ because they are subspaces of $V$. Thus the zero vector of $V$ is in $H \cap K$ . Let $\mathbf{u}$ and $\mathbf{v}$ be in $H \cap K$. Then $\mathbf{u}$ and $\mathbf{v}$ are in $H$. since $H$ is a subspace $\mathbf{u} + \mathbf{v}$ is in $H$. Likewise $u$ and $v$ are in $K$. since $K$ is a subspace $u + v$ is in $K$. Thus $u + v$ is in $H \cap K$. Let $u$ be in $H \cap K$ . Then $\mathbf{u}$ is in $H$. since $H$ is a subspace $c\mathbf{u}$ is in $H$. Likewise $\mathbf{v}$ is in $K$. since $K$ is a subspace $c\mathbf{u}$ is in $K$. Thus $c\mathbf{u}$ is in $H \cap K$ for any scalar $c$, and $H \cap K$ is a subspace of $V$.
The union of two subspaces is not in general a subspace. For an example in $\mathbb{R}^2$ let $H$ be the $x$ -axis and let $K$ be the $y$ -axis. Then both $H$ and $K$ are subspaces of $\mathbb{R}^2$, but $H \cup K$ is not closed under vector addition. The subset $H \cup K$ is thus not a subspace of $\mathbb{R}^2$ .

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# MTH007: Ch 4.2 Null space, column space and linear transformation

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1. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & 1 & 5 \end{bmatrix}.$$

(a) Find the reduced echelon form for the matrix $A$.

(b) Find a basis for $N(A)$ (the null space of $A$) and the dimension of $N(A)$.

(c) Find a basis for the column space of $A$ and a basis for the row space of $A$.

Solution:

(a) By performing elementary row operations, the reduced echelon form can be found as follows:

$$A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(b) Therefore a basis for the null space is $\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 0 \\ -2 \\ 1 \end{bmatrix}$ and the dimension of the null space is $\dim N(A) = 2$.

(c) Based on the reduced echelon form of $A$, $\text{rank}(A) = 2$ and the first and the third columns of $A$ are linearly independent. Therefore,

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} -1 \\ -3 \\ 1 \end{bmatrix}$$

form a basis for column space of $A$.

The row vectors

$$\begin{bmatrix} 1 & 2 & -1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 2 & 4 & -3 & 0 \end{bmatrix}$$

form a basis for the row space since elementary row operation does not change the row space. One can also use the row vector after row operations.

2. Let $a_1, a_2, a_3, a_4, a_5$ denote the columns of the matrix $A$, where

$$A = \begin{bmatrix} 5 & 1 & 2 & 2 & 0 \\ 3 & 3 & 2 & -1 & -12 \\ 8 & 4 & 4 & -5 & 12 \\ 2 & 1 & 1 & 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} a_1 & a_2 & a_4 \end{bmatrix}.$$

- Explain why $a_3, a_5$ are in the column space of $B$.
- Find a set of vectors that spans $\mathrm{Nul} A$ . one nor onto.

Solution:

a. To show that $\mathbf{a}_3$ and $\mathbf{a}_5$ are in the column space of $B$, we can row reduce the matrices $\begin{bmatrix} B & \mathbf{a}_3 \end{bmatrix}$ and

$$\begin{bmatrix} B & \mathbf{a}_5 \end{bmatrix} : \begin{bmatrix} B & \mathbf{a}_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1/3 \\ 0 & 1 & 0 & 1/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} B & \mathbf{a}_5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 10/3 \\ 0 & 1 & 0 & -26/3 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

since both these systems are consistent, $\mathbf{a}_3$ and $\mathbf{a}_5$ are in the column space of $B$. Notice that the same conclusions can be drawn by observing the reduced row echelon form for $A : A \sim \begin{bmatrix} 1 & 0 & 1/3 & 0 & 10/3 \\ 0 & 1 & 1/3 & 0 & -26/3 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

b. We find the general solution of $A\mathbf{x} = 0$ in terms of the free variables by using the reduced row echelon form of $A$ given above:
$x_1 = (-1/3)x_3 - (10/3)x_5, x_2 = (-1/3)x_3 + (26/3)x_5, x_4 = 4x_5$ with $x_3$ and $x_5$ free.

So $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -1/3 \\ -1/3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -10/3 \\ 26/3 \\ 0 \\ 4 \\ 1 \end{bmatrix}$, and a spanning set for Nul $A$ is

$$\left\{ \begin{bmatrix} -1/3 \\ -1/3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -10/3 \\ 26/3 \\ 0 \\ 4 \\ 1 \end{bmatrix} \right\}.$$

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# MTH007: Ch 4.3 Linearly independent Sets, Bases

**Year: 2022/23 Name: _____ ID: _____**

1. Suppose $\mathbb{R}^4 = \operatorname{span}\{v_1, v_2, v_3, v_4\}$. Explain why $\{v_1, v_2, v_3, v_4\}$ is a basis for $\mathbb{R}^4$.

Solution: Let $A = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 \end{bmatrix}$. Then $A$ is square and its columns $\operatorname{span}\mathbb{R}^4$ since $\mathbb{R}^4 = \operatorname{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$. So its columns are linearly independent by the Invertible Matrix Theorem, and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a basis for $\mathbb{R}^4$.

2. Let $H = \operatorname{span}\{u_1, u_2, u_3\}$ and $K = \operatorname{span}\{v_1, v_2, v_3\}$ where

$$u_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 3 \\ 4 \\ 1 \\ -4 \end{bmatrix} \text{ and } v_1 = \begin{bmatrix} -2 \\ -2 \\ -1 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 3 \\ 2 \\ -6 \end{bmatrix}, v_3 = \begin{bmatrix} -1 \\ 4 \\ 6 \\ -2 \end{bmatrix}$$

Find the base for $H, K$, and $H + K$.

Solution: Row reduce the following matrices to identify their pivot columns:

$$\begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 4 \\ 0 & -1 & 1 \\ -1 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \text{ so } \{\mathbf{u}_1, \mathbf{u}_2\} \text{ is a basis for } H.$$

$$\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} -2 & 2 & -1 \\ -2 & 3 & 4 \\ -1 & 2 & 6 \\ 3 & -6 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \text{ so } \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \text{ is a basis for } K.$$

$H + K$ is a subspace of $\mathbb{R}^4$ since it is closed under the operation of vector addition and scalar multipication and also includes $0$ vector since $0 = 0 + 0$ is in $H + K$. More details are as follows: For any two vectors $u$ and $v$ from $H + K$ and a scalar $\lambda \in \mathbb{R}$, there exist $h_1$ and $h_2$ in $H$ and $k_1$ and $k_2$ in $K$ such that $u = h_1 + k_1$ and $v = h_2 + k_2$. Then $u + v = (h_1 + k_1) + (h_2 + k_2) = (h_1 + h_2) + (k_1 + k_2)$ in $H + K$ and $\lambda(h_1 + k_1) = (\lambda h_1) + (\lambda h_2)$ in $H + K$ as both $H$ and $K$ are subspace of $\mathbb{R}^4$.

$$\begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 & -2 & 2 & -1 \\ 2 & 2 & 4 & -2 & 3 & 4 \\ 0 & -1 & 1 & -1 & 2 & 6 \\ -1 & 1 & -4 & 3 & -6 & -2 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 0 & 3 & -2 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \text{ so } \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{v}_2, \mathbf{v}_3\} \text{ is a basis for } H + K.$$

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# MTH007: Ch 4.6 Rank

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1. Let $A$ be any $2 \times 3$ matrix such that rank $A = 1$ , let $u$ be the first column of $A$, and suppose $u \neq 0$. Explain why there is a row vector $v$ in $\mathbb{R}^3$ such that $A = uv^T$. How could this construction be modified if the first column of A were zero?

Solution: Let $A = [u_1\ u_2\ u_3]$, and assume that rank $A = 1$. Suppose that $u_1 \neq \mathbf{0}$. Then $\{u_1\}$ is a basis for Col$A$, since Col$A$ is assumed to be one-dimensional. Thus there are scalars $x$ and $y$ with $u_2 = xu_1$ and $u_3 = yu_1$, and $A = u_1v^T$, where $v = \begin{bmatrix} 1 \\ x \\ y \end{bmatrix}$. If $u_1 = \mathbf{0}$ but $u_2 \neq \mathbf{0}$, then similarly $\{u_2\}$ is a basis for Col$A$, since Col$A$ is assumed to be one-dimensional. Thus there is a scalar $x$ with $u_3 = xu_2$, and $A = u_2v^T$, where $v = \begin{bmatrix} 0 \\ 1 \\ x \end{bmatrix}$.

If $u_1 = u_2 = \mathbf{0}$ but $u_3 \neq \mathbf{0}$, then $A = u_3v^T$, where $v = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

2. Let

$$A = \begin{bmatrix} 7 & -9 & -4 & 5 & 3 & -3 & -7 \\ -4 & 6 & 7 & -2 & -6 & -5 & 5 \\ 5 & -7 & -6 & 5 & -6 & 2 & 8 \\ -3 & 5 & 8 & -1 & -7 & -4 & 8 \\ 6 & -8 & -5 & 4 & 4 & 9 & 3 \end{bmatrix}$$

- Construct matrices $C$ and $N$ whose columns are basis for $\mathrm{Col}A$ and NulA, respectively, and construct a matrix $R$ whose rows form a basis for Row $A$.
- Construct a matrix $M$ whose columns form a basis for $\mathrm{Nul}A^T$ , form the matrices $S = \begin{bmatrix} R^T & N \end{bmatrix}$ and $T = [C \quad M]$, and explain why $S$ and $T$ should be square. Verify that $S$ and $T$ are invertible.

Solution:

(a) Begin by reducing $A$ to reduced echelon form:

$$A \sim \begin{bmatrix} 1 & 0 & 13/2 & 0 & 5 & 0 & -3 \\ 0 & 1 & 11/2 & 0 & 1/2 & 0 & 2 \\ 0 & 0 & 0 & 1 & -11/2 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

A basis for Col$A$ is the pivot columns of $A$, so matrix $C$ contains these columns:

$$C = \begin{bmatrix} 7 & -9 & 5 & -3 \\ -4 & 6 & -2 & -5 \\ 5 & -7 & 5 & 2 \\ -3 & 5 & -1 & -4 \\ 6 & -8 & 4 & 9 \end{bmatrix}.$$

A basis for Row$A$ is the pivot rows of the reduced echelon form of $A$, so matrix $R$ contains these rows:

$$R = \begin{bmatrix} 1 & 0 & 13/2 & 0 & 5 & 0 & -3 \\ 0 & 1 & 11/2 & 0 & 1/2 & 0 & 2 \\ 0 & 0 & 0 & 1 & -11/2 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

To find a basis for Nul$A$ row reduce to reduced echelon form, and the solution to $Ax = \mathbf{0}$ in terms of free variables is $x_1 = -(13/2)x_2 - 5x_5 + 3x_7$, $x_2 = -(11/2)x_3 - (1/2)x_5 - 2x_7$, $x_4 = (11/2)x_5 - 7x_7$, $x_6 = -x_7$, with $x_3, x_5$ and $x_7$ free. Thus matrix $N$ is

$$N = \begin{bmatrix} -13/2 & -5 & 3 \\ -11/2 & -1/2 & -2 \\ 1 & 0 & 0 \\ 0 & 11/2 & -7 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) The reduced echelon form of $A^T$ is

$$A^T \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -2/11 \\ 0 & 1 & 0 & 0 & -41/11 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 28/11 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

so the solution to $A^T x = \mathbf{0}$ in terms of free variables is $x_1 = (2/11)x_5$, $x_2 = (41/11)x_5$, $x_3 = 0$, $x_4 = -(28/11)x_5$, with $x_5$ free. Thus matrix $M$ is

$$M = \begin{bmatrix} 2/11 \\ 41/11 \\ 0 \\ -28/11 \\ 1 \end{bmatrix}.$$

The matrix $S = [R^T \ N]$ is $7 \times 7$ because the columns of $R^T$ and $N$ are in $\mathbb{R}^7$ and $dim\text{Row}A + dim\text{Nul}A = 7$. The matrix $T = [C \ M]$ is $5 \times 5$ because the columns of $C$ and $M$ are in $\mathbb{R}^5$ and $dim\text{Col}A + dim\text{Nul}A^T = 5$. Both $S$ and $T$ are invertible because their columns are linearly independent.

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# MTH007: Ch 5.1 Eigenvectors and Eigenvalues Assignment for Chapter 5

**Year: 2022/23 Name: _____ ID: _____**

1. For the matrix

$$A = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix},$$

find a basis for the eigenspace corresponding to eigenvalue $\lambda = 4$.

Solution: For $\lambda = 4$,

$$A - 4I = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 2 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Row reduce to $\begin{pmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. So $x_1 = 2x_3$, $x_2 = 3x_3$, with $x_3$ and $x_4$ free variables.

The general solution of $(A - 4I)x = \mathbf{0}$ is

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2x_3 \\ 3x_3 \\ x_3 \\ x_4 \end{pmatrix} = x_3 \begin{pmatrix} 2 \\ 3 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

A basis for the eigenspace is $\left\{ \begin{pmatrix} 2 \\ 3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$.

2. Without calculation, find one eigenvalue and two linearly independent eigenvectors of the matrix

$$A = \begin{pmatrix} 5 & 5 & 5 \\ 5 & 5 & 5 \\ 5 & 5 & 5 \end{pmatrix}.$$

Then justify your answer.

Solution: The matrix $A = \begin{pmatrix} 5 & 5 & 5 \\ 5 & 5 & 5 \\ 5 & 5 & 5 \end{pmatrix}$ is not invertible because its columns are linearly dependent. So the number $0$ is an eigenvalue of $A$. Eigenvectors for the eigenvalue $0$ are solutions of $Ax = \mathbf{0}$ and therefore have entries that produce a linear dependence relation among the columns of $A$. Any nonzero vector in $\mathbb{R}^3$ whose entries sum to $0$ will work. Find any two such vectors that are not multiples; for instance, $(1, 1, -2)$ and $(1, -1, 0)$.

3. Let $u$ and $v$ be eigenvectors of a matrix $A$, with corresponding eigenvalues $\lambda$ and $\mu$. Let $c_1$ and $c_2$ be scalars. Define $x_k = c_1\lambda^k u + c_2\mu^k v$, $(k = 0, 1, 2, \cdots)$.

   - Find $x_{k+1}$ by the definition;
   - Compute $Ax_k$ from the formula for $x_k$, and show that $Ax_k = x_{k+1}$.

Solution: a. Replace $k$ by $k+1$ in the definition of $x_k$, and obtain

$$x_{k+1} = c_1\lambda^{k+1}u + c_2\mu^{k+1}v.$$

b. Compute

$$\begin{aligned} Ax_k &= A(c_1\lambda^k u + c_2\mu^k v) \\ &= c_1\lambda^k Au + c_2\mu^k Av \quad \text{by linearity} \\ &= c_1\lambda^k \lambda u + c_2\mu^k \mu v \quad \text{since } u \text{ and } v \text{ are eigenvectors} \\ &= x_{k+1}. \end{aligned}$$

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# MTH007: Ch 5.2 The Characteristic Equation

**Year: 2022/23 Name: _____ ID: _____**

1. Find eigenvalues of matrix

$$A = \begin{pmatrix} 6 & -2 & 0 \\ -2 & 9 & 0 \\ 5 & 8 & 3 \end{pmatrix}$$

Solution: Make a cofactor expansion down the third column:

$$\det(A - \lambda I) = \det \begin{pmatrix} 6-\lambda & -2 & 0 \\ -2 & 9-\lambda & 0 \\ 5 & 8 & 3-\lambda \end{pmatrix} = (3-\lambda)\det \begin{pmatrix} 6-\lambda & -2 \\ -2 & 9-\lambda \end{pmatrix}$$

$$= (3-\lambda)[(6-\lambda)(9-\lambda) - (-2)(-2)] = (3-\lambda)(\lambda^2 - 15\lambda + 50)$$
$$= (3-\lambda)(\lambda-5)(\lambda-10) = -\lambda^3 + 18\lambda^2 - 95\lambda + 150.$$

2. It can be shown that the algebraic multiplicity of an eigenvalue $\lambda$ is always greater than or equal to the dimension of the eigenspace corresponding to $\lambda$. Find $h$ in the matrix $A$ below such that the eigenspace for $\lambda = 5$ is two dimensional.

$$A = \begin{pmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & h & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Solution: Row reduce the augmented matrix for the equation $(A - 5I)x = \mathbf{0}$:

$$\begin{pmatrix} 0 & -2 & 6 & -1 & 0 \\ 0 & -2 & h & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & -4 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & -2 & 6 & -1 & 0 \\ 0 & 0 & h-6 & 1 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & -3 & 0 & 0 \\ 0 & 0 & h-6 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

For a two-dimensional eigenspace, the system above needs two free variables. This happens if and only if $h = 6$.

3. Show that $A = QR$ with $Q$ invertible, then $A$ is similar to $B = RQ$.

Solution: If $A = QR$ with $Q$ invertible, and if $B = RQ$, then write

$$B = Q^{-1}QRQ = Q^{-1}AQ,$$

which shows that $B$ is similar to $A$.